

Madeleine E. Goertz

Curriculum Vitae

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EDUCATION

University of California, Berkeley, CA

Aug. 2025 – Present

Ph.D. in Mathematics

California Polytechnic State University, San Luis Obispo, CA

Sept. 2021 – June 2025

M.S. in Mathematics

GPA: 4.0/4.0

Advised by Patrick Orson, wrote master's thesis *On Diffeomorphism Groups of Surfaces*

B.S. in Mathematics

GPA: 4.0/4.0

Advised by Eric Brussel, wrote senior project *On Moduli Spaces of Triangles*

PREPRINTS & PUBLICATIONS

Journal Articles

1. The Torus of Triangles. (with Eric Brussel). (to appear in *Involve, A Journal of Mathematics*). arXiv:2303.11446.

Conference Proceedings

1. Exhaustive Generation of Pattern-Avoiding s -Words. (with Samuel Buick, Amos Lastmann, Kunal Pal, Helen Qian, Sam Tachen, Aaron Williams, Leah Williams, & Yulin Zhai). (to appear in *Proceedings of the 23rd International Conference on Permutation Patterns, 2025*). arXiv:2508.16039.

Preprints

2. The Quaternary Gray Code and How It Can Be Used to Solve Ziggurat and Other Ziggu Puzzles. (with Aaron Williams). arXiv:2411.19291. (preprint).
1. The Stack of Triangles Up to Similarity. (with Eric Brussel, Elijah Guptill, & Kelly Lyle). arXiv:2408.07792. (preprint, under review).

SELECTED AWARDS & SCHOLARSHIPS

Chancellor's Fellowship, UC Berkeley Spring 2026
Outstanding Master's Thesis Award, Cal Poly Math Department May 2026
Academic Excellence Award, Cal Poly Bailey College of Science and Math May 2025
Academic Excellence Award, Cal Poly Math Department May 2025
Charles J. Hanks Excellence in Mathematics Scholarship, Cal Poly Math Department May 2024
Frost Research Scholarship, Cal Poly Bailey College Sept. 2022 – June 2025
First place, Physical & Mathematical Sciences, 37th Annual CSU Student Research Competition Apr. 2023
Frost Summer Undergraduate Research Scholarship, Cal Poly Bailey College June 2022 & June 2023

RESEARCH PROGRAM PARTICIPATION

NSF SMALL REU, *Williams College* Summer 2024
Advised by Aaron Williams, on *Combinatorial Algorithms & Hamilton Paths*
Frost Summer Undergraduate Research Program, *Cal Poly* Summer 2023
Advised by Eric Brussel, on *Moduli Space of Triangles*
Frost Summer Undergraduate Research Program, *Cal Poly* Summer 2022
Advised by Eric Brussel, on *Poncelet's Theorem*

SELECTED RESEARCH COMMUNICATION

Talks for Mathematical Audiences

5. "The Quaternary Gray Code and Ziggu Puzzles." (with Aaron Williams). *FUN26: 13th International Conference on Fun with Algorithms*, Porquerolles, France, May 2026.

4. "Solving Ziggu Puzzles Using the Quaternary Gray Code." (with Aaron Williams). CM16 - Greedy Constructions of Gray Codes, *CanaDAM 2025: Canadian Discrete and Algorithmic Mathematics Conference*, Ottawa, Canada, May 2025. (virtual talk).
3. "Moduli Space of Triangles & Euclid's Similarity Theorems." (with Eric Brussel, Elijah Guptill, & Kelly Lyle). AMS Contributed Paper Session on Geometry, *Joint Mathematics Meetings*, San Francisco, CA. Jan. 2024.
2. "Geometry of the moduli spaces of similarity classes of triangles defined by three classical theorems." (with Eric Brussel, Elijah Guptill, & Kelly Lyle). *2nd CSU Mathematical Sciences Conference*, Bakersfield, CA. Nov. 2023.
1. "Poncelet Families & The Torus of Triangles." (with Eric Brussel). *1st CSU Mathematical Sciences Conference*, Northridge, CA. Nov. 2022.

Talks for General Audiences

3. "Solving Ziggu Puzzles Using the Quaternary Gray Code." (with Aaron Williams). *Cal Poly Bailey College of Science and Mathematics Student Research Conference*, San Luis Obispo, CA. May 2025.
2. "A Moduli Space of Triangles." (with Eric Brussel). *4th Annual CSU Grad Slam*, Virtual Meeting. May 2024.
1. "The Torus of Triangles." (with Eric Brussel). *37th Annual CSU Student Research Competition*, San Diego, CA. Apr. 2023.

Poster Presentations

2. "Three Moduli Spaces of Triangles." (with Eric Brussel, Elijah Guptill, & Kelly Lyle). *Bailey College of Science and Mathematics 2024 Student Research Conference*, San Luis Obispo, CA. May 2024.
1. "Poncelet Families & The Torus of Triangles." (with Eric Brussel). *Mathematics Department Summer Research Poster Symposium*, San Luis Obispo, CA. Oct. 2022.

TEACHING EXPERIENCE

Teaching experience as a graduate student instructor facilitating discussion sections at UC Berkeley.

Math 51: Calculus I, Fall Semester 2025

Teaching experience as a graduate teaching associate and instructor of record at Cal Poly.

Math 221: Calculus for Business and Economics, Spring Quarter 2025

Math 118: Precalculus Algebra, Fall Quarter 2024, Winter Quarter 2025

OTHER WORK EXPERIENCE

Mathematics Tutor, *Cal Poly Learning Support Center* Oct. 2023 – June 2024

Grader, *Cal Poly Department of Mathematics* Jan. 2023 – June 2023

Learning Assistant, *Cal Poly Department of Physics* Apr. 2022 – June 2022

Coding Camp Instructor, *Coding with Kids LLC* June 2019 – Sept. 2021